Classifying Motor Imagery from Open Source Datasets

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***Abstract*—The goal of this paper was to start learning about how BCI technology works and specifically how to use an EEG headset to predict motor imagery. The method that was attempted was to use the CSP method along with a Stockwell transform to process the data. A CNN was then used as the model to try to predict when a subject was thinking about performing movements with their left or right hand. The accuracy achieved was only 60% on the validation data which is much lower than expected. Hopefully this project will continue to be worked on to improve the accuracy and gain further knowledge in this emerging subject area.**

I. Introduction

A. Motivation

Brain-Computer Interface (BCI) technology is an exciting emerging field with applications from aiding those with disabilities to gaming. One of the most widely known groups developing this technology is Neuralink. Their method of getting data is by implanting electrodes in the brain [1]. However, this is an invasive medical procedure with numerous ethical concerns. Instead of that, the proposed approach was to use a traditional electroencephalogram (EEG) device to collect data and then predict the movement of a user's limbs. This is not a particularly groundbreaking idea, however processing EEG data using AI is a complex subject and this paper represents a first attempt at understanding and applying some common methodologies. Hopefully this paper will serve as a guide to someone interested in the field and eventually a marker of the progress that has been made.

B. Related Works

One resource that was found was a paper from 2022 by Chacon-Murguia and Rivas-Posada covering the classification of motor imagery (where the subject imagines performing a movement without actually performing any movement) using a Convolutional Neural Network (CNN) and a variety of processing methods. The accuracy they managed to achieve was between 86.47% and 97.67% depending on the processing methods used. [2] The system they propose is fast enough to classify 4 different classes of motor imagery in real time. CNNs are a common type of model used in this field and with the variety of methods used it was decided that a good first step would be to replicate this study.

C. Problem Definition

Originally the plan was to get access to an EEG device to generate data and then use that data to train and validate various methods. However, due to budget constrains access to an EEG device could not be obtained. Instead, publicly available datasets were used to prove the concept. Some early experimentation was done with the grasp-and-lift dataset [3] (Note: the dataset is currently hosted on kaggle), but the BCI-IV-2a dataset [4] was eventually settled on because it is clearly labelled and also used in many papers. This allowed for simpler comparisons to the results obtained in other papers.

The most accurate method from [2] was used. Which was to use Common Spatial Patterns (CSPs) and a Stockwell transform on the data and then use a CNN for classification.

II. Methodology

A. Data

The data obtained from the BCI-IV-2a dataset is composed of 22 channels of EEG data and 3 channels of electrooculogram (EOG) data stored in a .gdf file (General Data Format for Biomedical Signals) [4]. To generate the data 9 subjects were asked to imagine moving their left and right hands 72 times each, yielding 144 trials per subject. Note that the dataset includes two more classes of motor imagery but they were ignored to simplify the problem. Check the above citation for a full description. The method used only included 8 of the EEG channels. The MNE library handled many of the complex calculations needed and generated the visualizations [5].

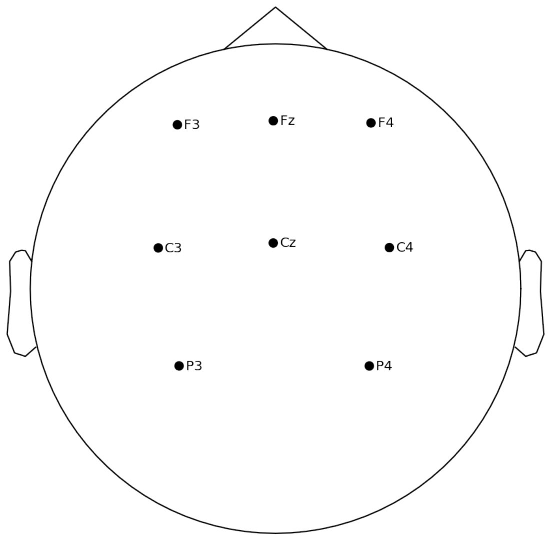

Fig. 1. A montage depicting the sensors used.

Figure 1 shows the which sensors were used and where they are located on the head. The labels correspond to the 10-20 system.

B. Filtering

A 7.5 to 30 Hz 4th order Butterworth band-pass filter was used to filter each channel because the alpha and beta waves (the brain waves in the 7.5 to 12 and 12 to 30 Hz ranges) are the ones relevant to motor imagery [2]. The following figures show the data before and after being filtered with dashed vertical lines representing 7.5 and 30 Hz.

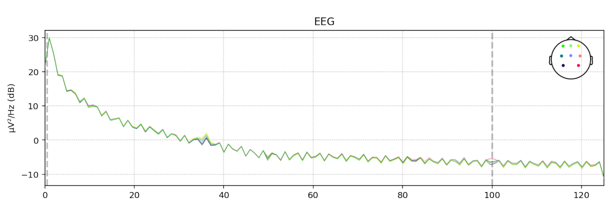

Fig. 2. A channel before being filtered.

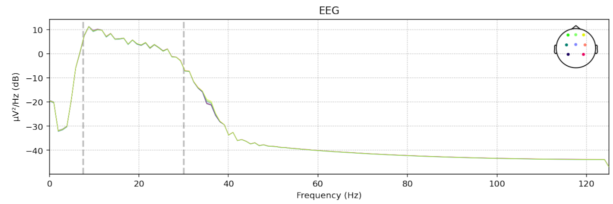

Fig. 3. A channel after being filtered.

C. Epoch extraction

From each trial in the dataset 2 seconds of activity were captured and labelled according to which action they were performing. At this point the trials involving feet and tongue motor imagery were discarded. The trial labels were also extracted and encoded into a one-hot encoding.

D. CSPs

CSPs use linear algebra to compute a matrix from the eigenvalues of the matrix of channels of the labelled data. Multiplying new data by this matrix will project it into a subspace of lower dimension. In this case it goes from having 8 channels to having just 2. The matrix will be unique to the subject but once the data from all the subjects is multiplied by the its subject's matrix it can all be used to train the CNN. The exact mathematical method is detailed in other papers which also give a better understanding of how this method works [6]. In the method used the CSP matrix was generated using the training data and then applied the matrix to the testing data

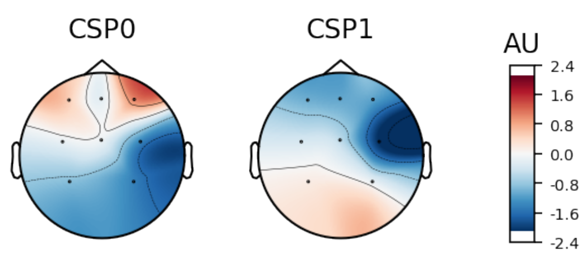

Fig. 4. A topographical representation of the CSP filter.

Figure 4 shows what regions of the brain tend to correlate with motor imagery. The red regions indicate that there is some correlation (whether positive or negative) where the blue regions indicate that there is typically no correlation. The units of the scale are arbitrary.

E. Stockwell transform

The Stockwell transform is a method of transforming the data from the time domain to the frequency domain. It is similar in function to a Fourier transform but the method is different.

Bicubic Interpolation

Depending on the parameters set for the CSPs and Stockwell transform the data may be either too large or in an awkward shape. Ideally the desired image is one that pooling can be applied to without shrinking it beyond the size of the kernel. Likewise, too large of an image would be require more memory and more computation time. So a bicubic interpolation was applied, which is a common method of shrinking images, to make the output into a 22x100 matrix.

F. CNN

The CNN used 3 convolutional layers of size 128 with 3x3 kernels and ReLU activations. After each convolutional layer there is a max pooling layer with a 2x2 pool size and a stride length of 1. After all of these layers the results are flattened into a fully-connected layer which feeds into a layer of 2 neurons with a softmax activation which represent the left and right hands.

III. RESULTS

The model was only able to achieve an accuracy of approximately 60% on the validation data. Unfortunately it overfits to the training data quite rapidly even when dropout layers are used. This is far below the expected performance. Reviewing the paper [2] the expected accuracy with only the CSP method is over 80%. This indicates that the CSP method used is likely flawed. Migrating away from the MNE library is a potential solution. However, the number of epochs extracted was also lower than the number extracted in the paper. For each epoch extracted they extracted 8 overlapping epochs which provided much more training data. MNE does not have an easy way of doing this so a custom epoch extraction method will likely have to be written.

IV. CONCLUSION

In this project some of the basic techniques that are used to process raw EEG data into data which is more comprehensible to a CNN were covered. Typically the process is to do filtering on the data, pass it through the CSP method and then do some form of transform from the time domain into the frequency domain. This paper used the Stockwell transform but the Continuous Wavelet Transform (CWT) or even a Short-Time Fourier Transform (STFT) could also work. The next steps are to try to get somewhere around an 80% accuracy using only the CSP method. Once that is achieved a transform can be applied to improve the accuracy further. Another option is do more research to find other sources and try to implement some of the techniques employed in them. Additionally, obtaining an EEG headset to record data is a goal for the future of the project. Once a grasp on the common methods used in this field has been attained more exploratory research could be done.

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